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Connector Stud, Connector Dock and Connector System

Abstract

A connector stud includes a head component including a head portion and a neck portion. The neck portion extends from a first side of the head portion. The head portion has a regular polygonal shape. The head component includes an annular groove in the first side of the head portion. The annular groove is configured to receive a protrusion of a connector dock.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to GB2403177.5 filed Mar. 5, 2024, the entire disclosure of

which is incorporated by reference.

FIELD

[0002] The present disclosure relates to a connector stud, connector dock and a connector system.

BACKGROUND

[0003] Connector systems are known for detachably connecting devices and/or equipment to a user's clothing or another item.

[0004] Such connector systems typically comprise a male connector stud which is attached to or incorporated into the equipment. A separate female dock may be located on the item to which the equipment is to be attached and the connector stud engages with the dock to affix the equipment in the required position.

[0005] Such connector systems may be used by personnel in a variety of sectors to connect and carry said devices and/or equipment. In many cases, the proper retention of the equipment is safety critical and thus the connector system must be sufficiently robust to avoid the equipment being inadvertently released from the connector dock.

[0006] The invention seeks to provide an improved connector stud and dock which has a low-profile design.

[0007] The background description provided here is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

SUMMARY

[0008] According to an aspect, there is provided a connector stud comprising: [0009] a head component comprising a head portion and a neck portion extending from a first side of the head portion, the head portion having a regular polygonal shape; [0010] wherein the head component comprises an annular groove in the first side of the head portion, the annular groove being configured to receive a protrusion of a connector dock.

[0011] The annular groove may encircle the neck portion.

[0012] The connector stud may further comprise a circular boss which protrudes from a second side of the head portion, the second side being opposed to the first side.

[0013] The connector stud may further comprise an exit slot which forms a discontinuity in a radially outer wall of the annular groove, the exit slot configured to allow withdrawal of the protrusion of the connector dock from the annular groove.

[0014] The connector stud may further comprise a base component comprising a base plate and a spacer which projects away from the base plate. At least one of the spacer and the neck portion may comprise a base attachment interface configured to attach the head component to the base component.

[0015] The base attachment interface may be formed by one of the spacer and the neck portion comprising a recess and the other of the spacer and the neck portion comprising a complementary protrusion which is received by the recess; wherein the recess and the protrusion are configured to prevent rotation of the head component relative to the base component. The base component may comprise a threaded insert which sits at least partially within a thickness of the spacer for receiving a screw which passes through the neck portion of the head component to fasten the head component to the base component.

[0016] According to another aspect, there is provided a device comprising a connector stud as described above.

[0017] The device may be an electronic device, such as a mobile communications device.

[0018] According to another aspect, there is provided a connector dock comprising: [0019] a receiving portion for receiving the head portion of the connector stud, the head portion having a regular polygonal shape; [0020] a cantilevered tab provided at a front of the receiving portion, the

tab having a protrusion which projects from a rear surface of the tab and extends towards a rear of the retaining portion; wherein the protrusion is configured to engage an annular groove formed in a first side of the head portion of the connector stud.

[0021] The receiving portion may further comprise a front curved surface which is configured to support a neck portion of the connector stud extending from the first side of the head portion.

[0022] The front curved surface may be formed by a pair of front shoulders.

[0023] The receiving portion may further comprise a rear curved surface which is configured to support a circular boss which protrudes from a second side of the head portion, the second side being opposed to the first side.

[0024] The rear curved surface is formed by a pair of rear shoulders.

[0025] The cantilevered tab may be disposed between the pair of front shoulders.

[0026] The receiving portion may comprise an opening forming a receiving section and a retaining section defined by a slot. The receiving portion may be configured to receive the head portion of the connector stud in a plane of the slot such that the connector stud can be translated to introduce the head portion to the slot.

[0027] The receiving portion may be provided with a rear flange which extends around a perimeter of the opening and is configured to limit axial movement of the connector stud relative to the connector dock.

[0028] The receiving portion may comprise an indexing mechanism which preferentially biases the head portion into a plurality of discrete angular positions.

[0029] The indexing mechanism may be formed by a pair of biasing surfaces which lie parallel to opposing sides of the head portion and are spaced from one another by a distance which is greater than the width between opposing sides of the head portion, but which is less than the width between opposing corners of the head portion.

[0030] The protrusion may be formed as a barb which has directional geometry. The protrusion may comprise an inclined inner surface and an upright outer surface.

[0031] According to another aspect, there is provided a connector system comprising a connector stud or device comprising a connector stud and a connector dock as described above.

[0032] The skilled person will appreciate that except where mutually exclusive, a feature described in relation to any one of the above aspects may be applied mutatis mutandis to any other aspect. Furthermore, except where mutually exclusive, any feature described herein may be applied to any aspect and/or combined with any other feature described herein.

[0033] Further areas of applicability of the present disclosure will become apparent from the detailed description, the claims, and the drawings. The detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] Embodiments will now be described by way of example only, with reference to the FIGS., in which:

[0035] FIG. 1 is a perspective view of an example connector stud according to the present disclosure;

[0036] FIG. 2 is an exploded view of the connector stud of FIG. 1;

[0037] FIG. 3 is a plan view of a base component of the connector stud;

[0038] FIG. 4 is a top perspective view of a head component of the connector stud;

[0039] FIG. 5 is a top view of the head component;

[0040] FIG. 6 is a bottom perspective view of the head component;

[0041] FIG. 7 is a bottom view of the head component;
[0042] FIG. 8 is a front perspective view of a connector dock for use with the connector stud;
[0043] FIG. 9 is a front view of the connector dock;
[0044] FIG. 10 is a rear perspective view of the connector dock;
[0045] FIG. 11 is a rear view of the connector dock;
[0046] FIGS. 12 to 17 show an example connector system according to the present disclosure, at different stages of an engagement action; and
[0047] FIGS. 18 to 22 show the connector system at different stages of a disengagement action.
[0048] In the drawings, reference numbers may be reused to identify similar and/or identical elements.

DETAILED DESCRIPTION

[0049] FIG. 1 shows an example connector stud 2 according to the present disclosure. The connector stud 2 is formed from a base component 4 and a head component 6. The base component 4 and the head component 6 are formed as discrete, separate elements which are detachably connected to one another using a fastener 8, such as a screw 8. However, in other examples, the base component 4 and the head component 6 may be integrally formed together.

[0050] Referring now to FIG. 2, which shows an exploded view of the connector stud 2, the base component 4 comprises a base plate 10 and a spacer 12 which is a circular protrusion (although other shapes may be used) which projects upwards from the base plate 10.

[0051] A non-circular recess 14 is formed at the center of the spacer 12. In this example, the recess 14 has a cruciform profile formed by four lobes, as best seen in FIG. 3. A threaded insert 16 is located at the center of the recess 14 within the body of the base plate 10. The threaded insert 16 sits at least partially within the thickness of the spacer 12.

[0052] With reference to FIGS. 4 to 7, the head component 6 comprises a head portion 18 and a neck portion 20 extending from a bottom side 23 of the head portion 18. The head portion 18 generally has a regular polygonal shape such that it has rotational symmetry. In this example, the head portion 18 has eight sides and so is octagon-shaped; however, it will be appreciated that other regular polygonal shapes may be used with fewer or greater sides.

[0053] A circular boss 22 protrudes from a top side 21 of the head portion 18, the top side 21 being opposed to the bottom side of the head portion 18. The circular boss 22 is concentric with the center (its axis of symmetry) of the head portion 18. The circular boss 22 has a radius which is less than the radius of the head portion 18 (i.e., the circumradius of the regular polygonal shape) and thus sits within an outer perimeter of the head portion 18.

[0054] As shown in FIGS. 6 and 7, a protrusion 24 extends from (or is formed by) the neck portion 20. The protrusion 24 is complementary to the recess 14. In this example, the protrusion 24 is formed by four separate protruding portions which are configured (i.e., are sized and positioned) to engage with the four lobes of the recess 14; although in other examples the protruding portions may be contiguous. The protrusion 24 and the recess 14 form part of a base attachment interface configured to attach the head component 6 to the base component 4.

[0055] The protrusion 24 of the head component 6 is engaged in the recess 14 of the base component 4 to prevent relative rotation between the head component 6 and the base component 4. The screw 8 is then passed through a hole 36 formed through the center of the circular boss 22, head portion 18 and neck portion 20 and engages with the threaded insert 16 to maintain the engagement between the protrusion 24 and the recess 14. The hole 36 is counterbored (or countersunk) such that the head of the screw 8 sits below the level of the circular boss 22. To aid assembly of the base component 4 and the head component 6, it is desirable that the protrusion 24 positively engages with the recess 14 so as to prevent axial movement of the protrusion 24 out of the recess 14 under a relatively small axial force. In particular, the protrusion 24 may clip into the recess 14 via a formation, such as a rib or ridge, or via an interference fit. This holds the head component 6 in place while the screw 8 is screwed into the base component 4.

[0056] An annular groove **26** is formed in the bottom side **23** of the head portion **18**. The annular groove **26** encircles the neck portion **20**. The annular groove **26** defines a radially inner wall **27** and a radially outer wall **29**. The radially inner wall **27** is spaced radially outward from the neck portion **20** and the radially outer wall **29** is spaced radially inward of the outer perimeter of the head portion **18**. In other examples, the radially inner wall **27** may be continuous with the neck portion **20** (i.e., the radially inner wall **27** of the annular groove **26** is formed by the neck portion **20**). The inner and outer walls **27**, **29** are substantially upright (vertical) and thus run parallel to one another.

[0057] An exit slot **30** is formed in the head portion **18**. The exit slot **30** forms a discontinuity in the radially outer wall **29** of the annular groove **26** and in the outer perimeter of the head portion **18**. The exit slot **30** is located centrally on one edge of the regular polygonal shape.

[0058] FIGS. **8** and **9** show a connector dock **37** for use with the connector stud **2** described previously. As shown, the connector dock **37** comprises a base plate **39** and a receiving portion **41** which projects upwards from the base plate **39**. The base plate **39** may be used to attach the connector dock **37** to clothing (e.g., by sewing or

[0059] The receiving portion **41** comprises an opening **38**. An upper (receiving) section of the opening **38** is provided with a rear flange **44** which extends around a perimeter of the opening **38** at the rear of the receiving portion **41** (in an axial direction).

[0060] A lower (retaining) section of the opening **38** is provided with a pair of front shoulders **46a**, **46b** which are spaced from one another at left and right sides of the opening **38** at the front of the receiving portion **41**. As shown in FIGS. **10** and **11**, a pair of rear shoulders **48a**, **48b** are also provided at the rear of the receiving portion **41** which are spaced from one another at left and right sides of the opening **38**. The respective front shoulders **46a**, **46b** are spaced from the respective rear shoulders **48a**, **48b** in an axial direction to define a pair of slots **50a**, **50b** therebetween. The slots **50a**, **50b** extend along the plane of the receiving portion **41**. As shown in FIG. **10**, the slots **50a**, **50b** are bounded by a pair of biasing surfaces **52a**, **52b** which face one another.

[0061] Each of the front shoulders **46a**, **46b** defines a curved surface. The curved surfaces may each define a circular arc which both lie on a common circle. Similarly, each of the rear shoulders **48a**, **48b** defines a curved surface. The curved surfaces may each define a circular arc which both lie on a common circle. The common circles defined by the front shoulders **46a**, **46b** and the rear shoulders **48a**, **48b** may have different radiuses but are concentric.

[0062] The receiving portion **41** further comprises a cantilevered tab **54** which is provided at the front of the receiving portion **41** between the front shoulders **46a**, **46b**. A retaining barb or protrusion **56** projects from a rear surface of the tab **54** and extends towards the rear of the retaining portion **41**. The barb **56** is located adjacent a distal end of the tab **54**.

[0063] The barb **56** has a directional geometry with an inclined inner surface **58** and an upright outer surface **60**.

[0064] The connector stud **2** and the connector dock **37** together form a connector system **100**. As shown in FIG. **12**, to connect the connector stud **2** to the connector dock **37**, the connector stud **2** is inserted into the upper receiving section of the opening **38**. The connector stud **2** is inserted such that the top side **21** of the head portion **18** engages the rear flange **44**, with the circular boss **22** disposed therebetween, as shown in FIG. **13**. The rear flange **44** therefore prevents further axial movement of the connector stud **2** relative to the connector dock **37** and positions the connector stud **2** in the plane of the slots **50a**, **50b**.

[0065] The connector stud **2** can then be translated along the opening **38** and into the lower retaining section. As shown in FIG. **14**, the head portion **18** is received in the slots **50a**, **50b** formed between the front shoulders **46a**, **46b** and the rear shoulders **48a**, **48b**.

[0066] The inclined inner surface **58** of the barb **56** rides over the bottom side **23** of the head portion **18** causing the tab **54** to be deflected forwards until the barb **56** is aligned with the annular groove **26**. The tab **54** is resiliently biased towards its rest position which causes the barb **56** to snap into the annular groove **26**, as shown in FIGS. **15** and **16**. The upright outer surface **60** of the

barb **56** engages the radially outer wall **29** of the annular groove **26** and thus prevents the connector stud **2** from being withdrawn from the connector dock **37**. This forms a locked configuration.

[0067] As shown in FIG. **15**, in this position, the neck portion **20** is received and supported by the curved surfaces of the front shoulders **46a**, **46b**. Similarly, as shown in FIG. **17**, the circular boss **22** is received and supported by the curved surfaces of the rear shoulders **48a**, **48b**.

[0068] As shown in FIG. **15**, the head portion **18** is received between the biasing surfaces **52a**, **52b**. The biasing surfaces **52a**, **52b** run parallel to one another and are spaced apart by a distance which corresponds to the width **w1** (as shown in FIG. **7**) between opposing sides of the head portion **18**. Specifically, the spacing between the biasing surfaces **52a**, **52b** is greater than the width **w1** between opposing sides of the head portion **18**, but less than the width **w2** between opposing corners (vertices) of the head portion **18** (i.e., the diameter of the circumcircle encircling the head portion **18**). Accordingly, rotation of the head portion **18** about its axis of symmetry is only possible by the deflection (i.e., deformation) of the biasing surfaces **52a**, **52b** or deformation of the head portion **18** itself. Such deformation must result in the spacing between the biasing surfaces **52a**, **52b** being greater than the diameter of the head portion **18** in order to allow rotation. It will be appreciated that after clearing opposing corners of the head portion **18**, the biasing surfaces **52a**, **52b** are allowed to return to their original rest position. The biasing surfaces **52a**, **52b** therefore form an indexing mechanism which preferentially biases the head portion **18** into a plurality of discrete angular positions in which opposing sides of the head portion **18** lie parallel with the biasing surfaces **52a**, **52b**. In this case, since the head portion **18** has eight sides, the connector stud **2** has seven preferred angular positions (at 45-degree intervals) with respect to the connector dock **37**.

[0069] In order to withdraw the connector stud **2** from the connector dock **37**, the connector stud **2** must first be rotated into an unlocked configuration. Specifically, the connector stud **2** must be rotated (in this case, by 180 degrees) such that the exit slot **30** is aligned with the barb **56**, as shown in FIGS. **18** and **19**. The exit slot **30** has a width which is greater than the width of the barb **56** and so the head portion **18** can be freely translated towards the upper receiving section of the opening **38** with the barb **56** being released from the annular groove **26**. Once the head portion **18** is fully located in the upper receiving section of the opening **38**, the connector stud **2** can be withdrawn from the opening **38**, as shown in FIG. **22**.

[0070] Locating the annular groove **26** on the bottom side **23** of the head portion **18** allows the thickness of the connector stud **2** and connector dock **37** to be reduced. This low-profile design reduces the likelihood of the connector stud **2** and connector dock **37** being inadvertently caught and thus reduces the likelihood of damage.

[0071] As described previously, the connector stud **2** utilizes a two-piece construction comprising the base component **4** and the separate head component **6**. This allows the head component **6** to be replaced if it breaks or becomes worn. Further, this allows the base component **4** and the head component **6** to be formed from dissimilar materials.

[0072] In particular, the head component **6** is formed from a self-lubricating material, such as Nylon 6. This allows the head component **6** of the connector stud **2** to rotate within the indexing mechanism formed by the connector dock **37**, while minimizing wear. The base component **4** may be connected to or integrally formed as part of the housing of a device or other equipment. The device may be an electronic device, such as a mobile communications device (e.g., a police radio). The base component **4** may therefore be formed from an impact resistant material, such as Acrylonitrile Butadiene Styrene (ABS).

[0073] As described previously, the non-circular recess **14** and corresponding protrusion **24** form an anti-rotation feature. Although non-circular, it is desirable for the recess **14** and protrusion **24** to have rotational symmetry in order to allow re-orientation of the head component **6** relative to the base component. In particular, in the example shown, where a cruciform recess **14** is used, the head component **6** can be reoriented through 90-degree intervals.

[0074] As described previously, the threaded insert **16** is received within the spacer **12** which also defines the recess **14** of the anti-rotation feature. Accordingly, the thickness of the connector stud **2** is minimized. The screw **8** which runs through the center of the neck portion **20** also provides reinforcement to the neck portion **20**.

[0075] In the example described above, the biasing surfaces **52a**, **52b** run parallel to one another to correspond to the opposing sides of the head portion **18**. In other examples, the head portion **18** may have an odd number of sides such that opposing sides are not parallel. In this case, the biasing surfaces **52a**, **52b** may be angled relative to one another to correspond with the sides of the head portion **18**.

[0076] Although it has been described that the head component **6** has the protrusion **24** and the base component **4** has the recess **14**, it will be appreciated that this arrangement may be reversed. Further, each of the head component **6** and the base component **4** may be provided with a combination of recesses and protrusions. Although, the recess **14** has been described previously as being non-circular, it will be appreciated that the anti-rotation feature could be formed by a plurality of circular recesses which together form a non-circular recess. In particular, the cruciform recess **14** shown could be replaced by four circular recesses which receive the four separate protruding portions of the neck portion **20**. In addition, although it has been described that the base attachment interface is formed by the recess **14** and the protrusion **24**, it will be appreciated that in other examples, the base attachment interface may be formed from any cooperating elements on the base component **4** and the head component **6** which enable the base component **4** to be attached to the head component **6** in a fixed manner.

[0077] It will be appreciated that the annular groove **26** may be formed as an annular sector and thus may not permit complete 360-degree rotation.

[0078] In other examples, the connector stud **2** may be removable from the connector dock **37** in any angular position. In such arrangements, the exit slot **30** is not required. To allow the barb **56** to be removed from the annular groove **26**, the barb **56** may have a unidirectional geometry formed by opposing inclined surfaces which allow the barb **56** to move into and out of the annular groove **26**. Alternatively, the radially outer wall **29** of the annular groove **26** may be inclined to allow the barb **56** to ride out of the annular groove **26**.

[0079] In other examples, the rear shoulders **48a**, **48b** may connect to one another to form a single curved surface for supporting the circular boss **22**. Similarly, the front shoulders **46a**, **46b** may connect to one another to form a single curved surface for supporting the neck portion **20**, with the cantilevered tab **54** located radially outward of the curved surface. With this arrangement the slots **50a**, **50b** may be connected to form a single slot for receiving the head portion **18** of the connector stud **2**.

[0080] In other examples, the head portion **18** may not be provided with the circular boss **22**. With such an arrangement, the opening **38** may be closed along a rear surface such that the slots **50a**, **50b** are formed between the front shoulders **46a**, **46b** and the rear surface (with the rear shoulders **48a**, **48b** no longer required).

[0081] It will be understood that the invention is not limited to the embodiments above-described and various modifications and improvements can be made without departing from the concepts described herein. Except where mutually exclusive, any of the features may be employed separately or in combination with any other features and the disclosure extends to and includes all combinations and sub-combinations of one or more features described herein.

[0082] The term “set” generally means a grouping of one or more elements. The elements of a set do not necessarily need to have any characteristics in common or otherwise belong together. The phrase “at least one of A, B, and C” should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.” The phrase “at least one of A, B, or C” should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR.

Claims

1. A connector stud comprising: a head component including a head portion and a neck portion extending from a first side of the head portion, the head portion having a regular polygonal shape, wherein the head component includes an annular groove in the first side of the head portion, the annular groove being configured to receive a protrusion of a connector dock.
2. The connector stud of claim 1, wherein the annular groove encircles the neck portion.
3. The connector stud of claim 1, further comprising a circular boss which protrudes from a second side of the head portion, the second side being opposed to the first side.
4. The connector stud of claim 1, further comprising an exit slot which forms a discontinuity in a radially outer wall of the annular groove, the exit slot configured to allow withdrawal of the protrusion of the connector dock from the annular groove.
5. The connector stud of claim 1, further comprising a base component including a base plate and a spacer which projects away from the base plate, wherein at least one of the spacer and the neck portion includes a base attachment interface configured to attach the head component to the base component.
6. The connector stud of claim 5, wherein: the base attachment interface is formed by one of the spacer and the neck portion including a recess, and the other of the spacer and the neck portion including a complementary protrusion which is received by the recess, the recess and the protrusion are configured to prevent rotation of the head component relative to the base component, and the base component includes a threaded insert which sits at least partially within a thickness of the spacer and is configured to receive a screw which passes through the neck portion of the head component to fasten the head component to the base component.
7. A device comprising the connector stud of claim 1.
8. The device of claim 7, wherein the device is an electronic device.
9. The device of claim 8, wherein the electronic device is a mobile communications device.
10. A connector dock comprising: a receiving portion configured to receive a head portion of a connector stud, the head portion having a regular polygonal shape; and a cantilevered tab provided at a front of the receiving portion, the tab having a protrusion which projects from a rear surface of the tab and extends towards a rear of a retaining portion, wherein the protrusion is configured to engage an annular groove formed in a first side of the head portion of the connector stud.
11. The connector dock of claim 10, wherein the receiving portion further includes a front curved surface which is configured to support a neck portion of the connector stud extending from the first side of the head portion.
12. The connector dock of claim 11, wherein the front curved surface is formed by a pair of front shoulders.
13. The connector dock of claim 10, wherein the receiving portion further includes a rear curved surface which is configured to support a circular boss which protrudes from a second side of the head portion, the second side being opposed to the first side.
14. The connector dock of claim 13, wherein the rear curved surface is formed by a pair of rear shoulders.
15. The connector dock of claim 14, wherein the cantilevered tab is disposed between the pair of front shoulders.
16. The connector dock of claim 10, wherein: the receiving portion includes an opening forming a receiving section and a retaining section defined by a slot, and the receiving portion is configured to receive the head portion of the connector stud in a plane of the slot such that the connector stud can be translated to introduce the head portion to the slot.
17. The connector dock of claim 16, wherein the receiving portion is provided with a rear flange which extends around a perimeter of the opening and is configured to limit axial movement of the

connector stud relative to the connector dock.

18. The connector dock of claim 10, wherein the receiving portion includes an indexing mechanism which preferentially biases the head portion into a plurality of discrete angular positions.

19. The connector dock of claim 18, wherein the indexing mechanism is formed by a pair of biasing surfaces which lie parallel to opposing sides of the head portion and are spaced from one another by a distance which is greater than the width between opposing sides of the head portion, but which is less than the width between opposing corners of the head portion.

20. The connector dock of claim 10, wherein the protrusion includes an inclined inner surface and an upright outer surface.

21. A connector system comprising: a connector stud including a head component including a head portion and a neck portion extending from a first side of the head portion, the head portion having a regular polygonal shape, wherein the head component includes an annular groove in the first side of the head portion, the annular groove being configured to receive a protrusion of a connector dock; and connector dock including a receiving portion and a cantilevered tab, the receiving portion configured to receive a head portion of the connector stud, the head portion having a regular polygonal shape, the cantilevered tab provided at a front of the receiving portion, the tab having a protrusion which projects from a rear surface of the tab and extends towards a rear of a retaining portion, wherein the protrusion is configured to engage an annular groove formed in a first side of the head portion of the connector stud.
